Design and Implementation of a MATLAB-Based Voice Recognition Anti-Theft System for Smart Parking Infrastructure

EEE 312: Digital Signal Processing I Laboratory — Section C1, Group 04

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***Abstract*—This paper presents a MATLAB voice-authentication system that releases a parked vehicle only after two independent spoken factors pass: a spoken roll number and full name (Factor 1), and a freshly generated random three-digit code (Factor 2). Each factor separates *what* was said, decided by dynamic time warping (DTW) on cepstral-normalised MFCC sequences, from *who* said it, decided by a Gaussian mixture model–universal background model (GMM-UBM). A single 3-s ambient-noise recording taken at the start of each login drives power spectral subtraction for every utterance in that login. With 29 enrolled students (493 recordings), held-out database trials rejected 0 of 348 genuine logins (95% upper bound 1.1%) and accepted 0 of 87 unregistered speakers who said their own details (upper bound 4.2%); the phrase gate alone has an equal error rate (EER) of at most 1.1%. In 29 live sessions on 30 September 2026, 5 were granted, 12 stopped at Factor 1 and 12 at Factor 2, half of the latter at the challenge voice check alone. Compared with the previous batch's MFCC + vector quantisation (VQ) system, the proposed system adds spoken-content verification, a replay-resistant challenge, integrated noise reduction and a held-out decision threshold, and removes the PIN path that let a user bypass voice verification. A fail-closed hardware integration, in which an ESP32 controller moves a barrier only on an authenticated grant message from MATLAB, is also proposed in Appendix B but has not yet been built.**

***Index Terms*—challenge–response, cohort normalisation, dynamic time warping, GMM-UBM, MAP adaptation, MFCC, spectral subtraction, speaker verification.**

I. Introduction

Vehicle theft from parking facilities usually exploits a transferable token: a ticket, key card or PIN that works for whoever holds it. A voice is harder to hand over, so it is a natural second line of defence at the point where a vehicle is released.

Voice authentication has two well-known failure modes. A fixed spoken password can be secretly recorded and replayed, and background noise in a garage changes the features the system relies on. The previous batch's campus bicycle system showed both: it used a fixed name-and-ID phrase, and its accuracy fell from 97.5% on clean recordings to 73.81% in real use [1].

Demonstration video: https://youtu.be/20vIgqYNTaI

A. Objectives

1) Verify a claimed identity from speech alone, with no PIN or card fallback that bypasses the voice check.
2) Verify both what was said (the right roll number and name) and who said it (the enrolled owner of that identity).
3) Defeat simple replay of a recorded login by requiring a fresh, unpredictable spoken response.
4) Reduce the effect of steady room noise using a live estimate of that noise.
5) Learn decision thresholds from data and report error rates, with confidence intervals, measured on recordings not used to build the models.
6) Specify a fail-closed path from the software decision to a physical barrier.

B. Contributions

- A two-factor login: roll number + name (Factor 1), then a random three-digit challenge (Factor 2), each with separate content and speaker checks.
- Content matching by DTW on time-ordered MFCC sequences, replacing the previous batch's order-free VQ codebooks.
- Speaker verification by GMM-UBM with MAP adaptation, with a dedicated digit-only model for the challenge.
- One 3-s ambient-noise reference per login, shared by all three utterances.
- A phrase threshold learned by five-fold "unregistered speaker" cross-validation, with held-out FRR/FAR, exact

95% confidence intervals, EER bounds and stage-wise live pass rates (Section VII).
- A proposed (unbuilt) hardware design, given in Appendix B, with a hardware random-number source, an authenticated grant message and fail-closed barrier logic.

The implemented system is software-only and runs in MATLAB without the Signal Processing, Statistics or Image Processing toolboxes. The hardware in Appendix B is a design that has not been built or tested; no hardware result is claimed.

II. Literature Review

The system combines six established techniques; the contribution is how they are arranged and calibrated, not any new algorithm.

Mel-frequency cepstral coefficients (MFCC): Davis and Mermelstein showed that a mel-spaced filterbank, a log and a DCT give compact features that outperform linear-prediction features for word recognition [2]. First-order regression ("delta") coefficients add the short-term trajectory of the spectrum [3].

Vector quantisation (VQ): The LBG algorithm builds a small codebook of spectral centroids per speaker [4]. A test utterance is scored by its average distance to the nearest codeword. VQ is compact and fast, but it ignores the order of frames, so it cannot tell which words were spoken. The previous batch used this method [1].

Dynamic time warping (DTW): DTW finds the lowest-cost monotonic alignment between two feature sequences [5]. It absorbs differences in speaking rate while keeping temporal order, which makes it a strong template matcher for isolated words and short fixed phrases.

GMM-UBM speaker verification: Reynolds *et al.* model "speech in general" with one large Gaussian mixture (the UBM). Each speaker's model is the UBM with its means MAP-adapted towards that speaker's frames [6]. The score is a log-likelihood ratio (LLR) of speaker model against UBM, which cancels much of the channel and phrase variation.

Score normalisation: Raw distances drift with microphone, room and loudness. Cohort normalisation divides by the distance to the closest competing speakers, so a common shift cancels [7]. Cepstral mean and variance normalisation (CMVN) removes channel colouration from content features [8].

Spectral subtraction: Boll subtracts an estimated noise power spectrum from each short-time frame [9]. Berouti *et al.* added over-subtraction (α) and a spectral floor (β) to suppress "musical noise" [10].

Replay and challenge–response: A fixed pass-phrase can be recorded and replayed. Text-prompted verification asks for content the attacker cannot predict, so a pre-recorded login no longer suffices [11]. It does not, by itself, prove that a live person is speaking.

Performance evaluation: Biometric verification is reported with false acceptance and false rejection rates at a stated operating point, the equal error rate, and the detection error trade-off (DET) curve [12], [13]. Small trial counts call for confidence intervals; for zero observed errors the exact Clopper–Pearson interval [14] gives a meaningful upper bound where a point estimate of 0% does not.

III. Baseline: The Previous Batch's System

The previous batch built "Secured Campus Transportation System Using Voice-Based Authentication", a MATLAB GUI for bicycle check-in and check-out [1]. A user entered an ID, spoke a fixed name and ID, and was matched by MFCC features against per-speaker VQ codebooks.

Pipeline: Pre-emphasis ($\alpha = 0.97$), short-time-energy silence removal, 25-ms framing, a mel filterbank from 300 Hz, DCT, and static + delta + delta-delta coefficients. A 32-codeword LBG codebook was trained per speaker, and the speaker with the minimum average Euclidean distortion was taken as the match. After three voice failures, a 4-digit PIN alone could grant access.

Reported results: 78 of 80 clean test utterances (16 speakers) were identified correctly (97.5%). In real, noisy use, 31 of 42 login attempts succeeded (73.81%). Impostor rejection was described qualitatively, without a false-acceptance figure.

Limitations we set out to fix:

- *Replay:* the only spoken input was a fixed phrase, so a recording of the owner could be replayed.
- *Voice bypass:* the PIN fallback let anyone who learned the PIN skip the biometric check entirely.
- *No content check:* VQ pools frames without order; it reports whose voice a recording resembles, not whether the correct ID was spoken.
- *Noise sensitivity:* the 24-point drop from clean to real use shows the pipeline had no effective noise compensation.
- *Closed-set evaluation:* "nearest speaker" accuracy always names someone; it does not measure how often an outsider is wrongly accepted, so no FAR or FRR was reported.

IV. System Design

A login is a fixed sequence of seven checks; the first failure ends it with ACCESS DENIED, and there is no retry or fallback inside the same login (Fig. 1).

The random code is generated only after Factor 1 has confirmed the claimed student, so it is never shown to an unverified speaker. Both factors split the question into what was said (DTW) and who said it (GMM-UBM), because the two need different features and different normalisation.

V. Methodology

Every recording passes through the same front end; Factor 1 and Factor 2 then apply separate content and speaker tests. All processing is at 8 kHz, and live audio is held only in memory. Framing, windowing, the DFT and FIR pre-emphasis follow [15], [16].

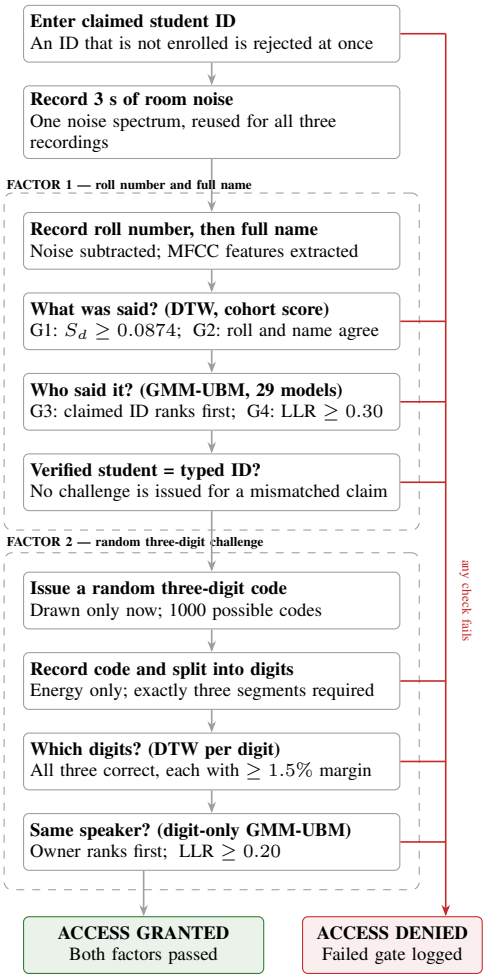


Fig. 1. Live login flow: two factors, seven checks. Any failed check ends the login.

TABLE I
THREAT MODEL

Attacker	Example	Main defence
Unenrolled outsider	Says a valid roll number and name	G1 phrase gate and G4 voice level; the outsider has no model of their own to win G3
Enrolled classmate	Says a friend's roll number and name	G3: the classmate's own model fits their voice better than the friend's
Replay	Plays a recording of the owner's roll number and name	Factor 2: a fresh code in the owner's voice
Wrong claim	Types one ID but says another	Claim check before the challenge
Link forgery*	Sends a fake "open" command to the barrier controller	HMAC-signed, single-use grant (Appendix B)

*Applies only to the proposed hardware. Splicing the owner's recorded digits, synthetic speech and tampering with the computer itself are outside this threat model (Section IX).

A. Enrollment Data

Each student has one folder of WAV files named `<roll>_<utterance>_<take>.wav`. A complete set is 17 files: 3 roll-number takes, 3 full-name takes, one 0–9 digit sequence, and one isolated take of each digit 0–9. The database holds 29 students and 493 files, about 16 minutes of speech after trimming. A roll number lasts 5.3 s (median) and an isolated digit 0.5 s.

B. Pre-Processing

1) Convert to mono, resample to 8 kHz with an anti-aliasing resampler, and remove the DC offset.
2) Apply power spectral subtraction (Section V-C).
3) Normalise to unit peak, unless the recording is near-silent (RMS < 0.006 and peak < 0.025).
4) Trim leading and trailing silence by short-time energy: 20-ms windows, 10-ms hop, threshold 1.5% of the peak frame energy, with a two-frame guard band.
5) Pre-emphasise with (1) and frame with a 25-ms Hamming window at 50% overlap (200 samples, 100-sample hop).

$$y[n] = x[n] - 0.95\,x[n-1], \qquad H(z) = 1 - 0.95\,z^{-1} \quad (1)$$

C. Ambient-Noise Reduction

Before the user speaks, the system records 3 s of room sound. It is accepted only if it is at least 2.75 s long, unclipped, between -85 and -20 dBFS, and free of speech bursts. Its mean power spectrum $P_n(k)$ is then computed with the same 32-ms periodic-Hann STFT used for subtraction. That one profile is reused for the roll number, the name and the challenge. If it fails a check, each utterance falls back to the mean of its own quietest 10% of frames; enrollment always uses this fallback.

Each STFT frame (256 samples, 50% overlap) is scaled by a spectral gain and rebuilt by overlap-add, keeping the noisy phase [9], [10]:

$$G(k) = \max\left(1 - \alpha\,\frac{P_n(k)}{|X(k)|^2},\ \beta\right), \quad (2)$$

and the enhanced spectrum is $Y(k) = X(k)\sqrt{G(k)}$, with $\alpha = 2$ and $\beta = 0.02$.

D. Feature Extraction

Each frame's 512-point power spectrum passes through 26 triangular mel filters from 100 to 4000 Hz, then a log and an orthonormal DCT-II. The energy term c_0 is dropped. The mel scale and delta coefficients are

$$m(f) = 2595\log_{10}\left(1 + \frac{f}{700}\right), \quad (3)$$

$$\Delta c_t = \frac{\sum_{n=1}^{2} n\,(c_{t+n} - c_{t-n})}{2\sum_{n=1}^{2} n^2}. \quad (4)$$

Two feature sets are built from the same filterbank, because content and identity need different normalisation (Table II). Only speech frames are kept: those whose total mel log-energy is within 30 dB of the loudest frame. This also removes pauses between the digits of a roll number.

TABLE II
FEATURE SETS DERIVED FROM ONE FILTERBANK

Use	Coefficients	Normalisation	Dim.
Content (what was said)	c_1–c_{10} + Δ	CMVN along time, per utterance or per digit	20
Speaker (who said it)	c_1–c_{12} + Δ	Cepstral mean subtraction only	24

TABLE III
FACTOR 1 DECISION GATES

Gate	Condition	What it stops
G1 Phrase	$S_d(c) \geq \theta_{\text{phrase}} = 0.0874$	Phrases that are not a registered roll number and name
G2 Consistent	Roll and name both point to c	Mixing one student's roll with another's name
G3 Voice rank	c has the highest LLR of all 29 students	A classmate saying someone else's details
G4 Voice level	$\text{LLR}(c) \geq 0.30$	An unenrolled speaker, who has no model to win G3

E. Factor 1: Roll Number and Name

Content: For each utterance u and student c, $d(u, c)$ is the lowest DTW cost between u and any of c's takes of the same phrase. Local cost is Euclidean distance and the total is divided by the warping-path length, so long and short phrases are comparable:

$$D(i,j) = \|\mathbf{a}_i - \mathbf{b}_j\| + \min\{D(i\text{–}1, j),\, D(i, j\text{–}1),\, D(i\text{–}1, j\text{–}1)\}, \tag{5}$$

$$d = D(T_a, T_b)/L_{\text{path}}. \tag{6}$$

The distance is cohort-normalised against the five closest other students [7], then averaged over roll and name:

$$S_d(u,c) = \log \frac{\text{mean}_5\{d(u,j) : j \neq c\}}{d(u,c)}, \tag{7}$$

$$S_d(c) = \tfrac{1}{2}\left[S_d(\text{roll}, c) + S_d(\text{name}, c)\right]. \tag{8}$$

The claimed student is the one with the highest S_d.

Speaker: A 128-component diagonal-covariance UBM is trained by EM (20 iterations, LBG initialisation) on the speaker frames of all students. Each student's model adapts only the UBM means, with relevance factor $r = 16$ [6]. The score is the average per-frame LLR over both utterances:

$$\boldsymbol{\mu}_k^{(c)} = a_k E_k[\mathbf{x}] + (1 - a_k)\,\boldsymbol{\mu}_k^{\text{UBM}}, \qquad a_k = \frac{n_k}{n_k + r}, \tag{9}$$

$$\text{LLR}(c) = \frac{1}{T}\sum_{t=1}^{T}\left[\log p(\mathbf{x}_t \mid c) - \log p(\mathbf{x}_t \mid \text{UBM})\right]. \tag{10}$$

Decision: Factor 1 passes only if all four gates in Table III pass for the claimed student c, and c equals the ID the user typed.

F. Factor 2: Random Three-Digit Challenge

Only after Factor 1 passes does the system draw a code of three digits, each uniform on 0–9 (1000 possible codes). The user speaks the digits with short pauses in one recording.

Segmentation: The noise-reduced signal is split by energy alone, never using the expected digits. A 20-ms RMS envelope (10-ms hop) is smoothed and thresholded at $\max(lo + 0.16(hi - lo),\, 0.085\, hi)$, where lo and hi are its 15th and 95th percentiles. Dips under 0.12 s are bridged, islands under 0.11 s are dropped, and gaps under 0.16 s are merged. Exactly three segments must remain, or the attempt is rejected.

Content: Each segment is compared by DTW with up to three takes of every digit 0–9 from up to six template speakers, the claimed owner first. A digit's score is $0.6\times$ the owner's distance $+\,0.4\times$ the mean of the best two speakers. The lowest-scoring digit wins, and it must beat the runner-up by a relative margin of at least 1.5%. All three digits must be correct and confident.

Speaker: Scoring uses a separate digit-only GMM-UBM (128 components), trained only on the isolated digit recordings, at most 1200 frames per student. Isolated digits are short (median 0.5 s) and acoustically unlike the long phrases the Factor 1 model was trained on, so they get their own model. The claimed owner must rank first among all 29 students, with an LLR of at least 0.20.

G. Learning the Phrase Threshold

θ_{phrase} is learned so that an unregistered speaker's phrase rarely passes G1. The 87 roll + name pairs are split into five folds by student. In each fold, about six students are removed from the enrolled set and the UBM is refit without them; their logins are then scored as strangers. θ_{phrase} is the 98th percentile of their best S_d (target 2% false acceptance at G1), giving 0.0874.

The voice gates are set by hand rather than learned. All enrollment takes of a student come from one sitting, so held-out database LLRs (5th percentile 0.97) are far higher than live logins on another day (0.39–1.21 for passing live sessions). A threshold fitted to the database would reject most real users (quantified in Section VII-C). A supervised calibration script, `CALIBRATE_FACTOR2.m`, can set a per-student Factor 2 gate from 12 genuine and 12 impostor live trials, but no calibration has been applied yet.

VI. RESULTS

On held-out database trials Factor 1 made no errors except against an outsider who knows whom they sound like; in live use, the Factor 2 voice check was the main reason sessions that had passed Factor 1 were denied.

A. Enrolled Models

The database, rebuilt on 30 September 2026, holds 29 students, each with all 17 recordings. The Factor 1 UBM was trained on 42 154 speech frames. The digit-only UBM was trained on 7 696 frames, 185–399 per student, which is only 1.9–4.0 s of speech each.

TABLE IV
HELD-OUT DATABASE EVALUATION

Measure	Result
Phrase points to the correct student	100% (348/348)
Voice ranks the correct student first	100% (348/348)
Genuine logins rejected (FRR)	0.0% (0/348)
Unregistered speaker, own roll + name, accepted	0.0% (0/87)
Registered classmate claiming another student, accepted	0.0%
Unregistered speaker saying details of closest-voice student, accepted (worst case)	27.6% (24/87)
Phrase threshold θ_{phrase}	0.0874
Phrase score S_d: genuine median / highest unregistered	0.430 / 0.0905
Voice LLR: genuine median / 5th percentile	1.39 / 0.97
Voice LLR, closest-voice unregistered: median / maximum	0.15 / 0.80

TABLE V
LIVE SESSIONS BY STAGE REACHED (30 SEPTEMBER 2026)

Stage reached	Sessions	Denied here
Login started	29	—
Factor 1 passed	17	12
Challenge split into exactly 3 digits	15	2
All 3 digits correct and confident	11	4
Challenge voice passed (ACCESS GRANTED)	5	6

B. Held-Out Evaluation on the Database

The protocol of Section V-G gives 348 genuine logins (each student scored with their own model re-adapted without the test pair) and 87 unregistered-speaker logins. Results are in Table IV.

The phrase gate separates genuine and unregistered logins with a wide margin: only 1 of 87 unregistered logins (1.1%) scored above θ_{phrase}, within the 2% target. The 27.6% worst case is Factor 1 alone: it assumes an outsider has found the student whose voice is nearest theirs and knows that student's roll number and name; Factor 2 must then still be passed. These are same-session figures. Every take of a student was recorded in one sitting, so they overstate live performance.

C. Live Sessions

Twenty-nine live GUI sessions were logged on 30 September 2026. Five ended in ACCESS GRANTED and 24 in ACCESS DENIED (Table V, Fig. 2). The log records system decisions, not who was actually speaking, so it cannot give a true FRR or FAR. Three changes were made during the day: the database was rebuilt at 13:54, θ_{phrase} changed from 0.0795 to 0.0874, and the Factor 2 gate changed from 0.22 to 0.20 at 14:24.

Factor 1: 10 of the 12 Factor 1 denials failed G3: another enrolled student's voice model fitted better than the claimed student's. In one session the speaker typed 2206154, but the phrase and voice both identified 2206143 (LLR 1.36, rank 1); the claim mismatch was rejected before any challenge was issued. In passing sessions the Factor 1 LLR ranged from 0.39 to 1.21, below the held-out median of 1.39.

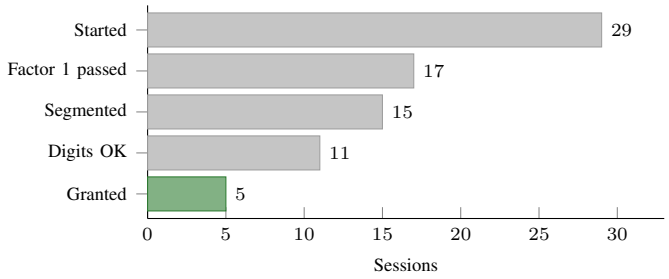


Fig. 2. Live-session funnel: sessions remaining after each stage (29 sessions).

Digit recognition: Across the 15 segmented challenges, 42 of 45 digits (93%) and 13 of 15 full codes were recognised correctly. Both segmentation failures produced 2 or 4 speech islands instead of 3, because pauses between digits were too short or a digit was split.

Challenge voice: In the 5 granted sessions the owner ranked first with LLR 0.29–0.71. In the 6 sessions denied only by the voice check, the digits were right but the owner ranked 3rd to 15th, with LLR -0.06 to 0.13. One student passed Factor 1 six times in 10 minutes and was denied by the challenge voice check in four of them. The digit-only model, trained on 2–4 s of speech per student, is the weakest link.

VII. PERFORMANCE ANALYSIS

This section restates the measured outcomes as standard verification metrics, attaches uncertainty to them, and lists the metrics that still need new measurements. All values below are derived from the counts and score statistics in Section VI; no new experiment was run.

A. Metric Definitions

For a decision threshold θ applied to a score s,

$$\text{FAR}(\theta) = \frac{\#\{\text{impostor trials with } s \geq \theta\}}{\#\{\text{impostor trials}\}}, \tag{11}$$

$$\text{FRR}(\theta) = \frac{\#\{\text{genuine trials with } s < \theta\}}{\#\{\text{genuine trials}\}}. \tag{12}$$

The EER is the error rate at the threshold where FAR and FRR are equal, and the half total error rate is $\text{HTER} = (\text{FAR} + \text{FRR})/2$ at the chosen operating point [13]. For k errors in n trials we report the exact two-sided 95% Clopper–Pearson interval [14],

$$\left[\, B^{-1}(0.025;\, k,\, n-k+1)\,,\;\; B^{-1}(0.975;\, k+1,\, n-k)\right], \tag{13}$$

where $B^{-1}(\cdot; a, b)$ is the inverse beta CDF, with the lower limit 0 when $k = 0$ and the upper limit 1 when $k = n$.

TABLE VI
HELD-OUT VERIFICATION METRICS WITH EXACT 95% CONFIDENCE INTERVALS

Metric	k/n	Est.	95% CI
FRR, genuine logins	0/348	0.0%	0–1.1%
FAR, unregistered, own details	0/87	0.0%	0–4.2%
G1 false pass, unregistered	1/87	1.1%	0.03–6.2%
FAR, informed outsider (Factor 1 only)	24/87	27.6%	18.5–38.2%
Phrase identification accuracy	348/348	100%	98.9–100%
Voice rank-1 identification accuracy	348/348	100%	98.9–100%

B. Held-Out Error Rates With Confidence Intervals

Table VI shows that the zero-error results are strong but not unlimited: with 348 genuine trials the FRR is below 1.1% at 95% confidence, and with 87 unregistered trials the FAR is below 4.2%. Proving a FAR below 1% at the same confidence would need about 370 error-free impostor trials. At the operating point, HTER is 0% for unregistered speakers and 13.8% for the informed-outsider case (FAR 27.6%, FRR 0%), the latter for Factor 1 alone.

C. EER Bounds and Operating Margins

The raw score vectors are not reproduced in this report, so the exact EER and DET curve are not given. A bound follows from the reported operating points. Because FAR(θ) is non-increasing and FRR(θ) is non-decreasing in θ, for any threshold θ_0,

$$\text{EER} \le \max\{\text{FAR}(\theta_0), \text{FRR}(\theta_0)\}. \tag{14}$$

Phrase score (G1): At $\theta_0 = 0.0874$, FRR $= 0/348$ and FAR $= 1/87$, so $\text{EER}_{\text{phrase}} \le 1.1\%$. The genuine median (0.430) is $4.9\times$ the threshold.

Voice LLR, informed outsider: The highest closest-voice LLR is 0.80 and the genuine 5th percentile is 0.97. Any θ_0 between them gives FAR $= 0/87$ and FRR $< 5\%$, so $\text{EER}_{\text{voice}} \le 5\%$ on database trials; the genuine and closest-voice medians differ by 1.24.

Usability trade-off: Moving G4 from 0.30 to a value in that interval (e.g., 0.85) would remove the 27.6% worst case on held-out data. It would also have rejected every live session that passed Factor 1 with an LLR between 0.39 and 0.85. The hand-set 0.30 therefore deliberately trades informed-outsider resistance for live usability until multi-session enrollment narrows the gap between database and live scores.

D. Live Pipeline Metrics

Table VII gives conditional pass rates for each stage, i.e., the fraction of sessions entering a stage that passed it. Factor 2 passed only 5 of the 17 sessions that cleared Factor 1 (29.4%, CI 10.3–56.0%), and within Factor 2 the voice check was the lowest-rate stage. Two of the 13 correctly recognised codes were rejected only by the 1.5% confidence margin, which is a tunable source of false rejections. Because the log has no ground-truth labels, these are pass rates, not FRR or FAR.

TABLE VII
LIVE STAGE METRICS WITH EXACT 95% CONFIDENCE INTERVALS

Stage / metric	k/n	Rate	95% CI
Factor 1 pass	17/29	58.6%	38.9–76.5%
Factor 1 denials at G3	10/12	83.3%	51.6–97.9%
Segmentation (exactly 3 digits)	15/17	88.2%	63.6–98.5%
Digit content (correct and confident)	11/15	73.3%	44.9–92.2%
Per-digit recognition accuracy	42/45	93.3%	81.7–98.6%
Full-code recognition accuracy	13/15	86.7%	59.5–98.3%
Challenge voice pass	5/11	45.5%	16.7–76.6%
Factor 2 pass, given Factor 1 passed	5/17	29.4%	10.3–56.0%
Overall grant rate	5/29	17.2%	5.8–35.8%

TABLE VIII
METRICS NOT YET MEASURED AND HOW TO OBTAIN THEM

Metric	Required data and procedure
Exact EER, DET curve	Export genuine and impostor S_d and LLR vectors from the held-out script; sweep θ and plot on normal-deviate axes [12]
Live FAR and FRR	Labelled multi-day study: genuine users, classmates, outsiders and replays, with the true speaker recorded per session
Latency per login	`tic`/`toc` around each stage over at least 30 logins; report mean and maximum
Accuracy vs. SNR	Mix recorded garage noise at 20, 10, 5 and 0 dB into clean test logins; compare with and without subtraction
Replay acceptance	Play recorded logins through a loudspeaker at the microphone; count Factor 1 and Factor 2 passes

E. Challenge Strength

A uniformly drawn three-digit code is guessed with probability 10^{-3} per attempt, so a replayed Factor 1 recording plus a guessed answer succeeds at most once in 1000 attempts even before the Factor 2 voice check. Because each login has a single attempt and no retry, an attacker must repeat Factor 1 for every guess. This bound does not cover splicing, where an attacker holds the owner's recordings of all ten digits (Section IX).

F. Metrics Still to Be Measured

Table VIII lists the metrics that need data this project has not yet collected. They are listed rather than estimated.

VIII. COMPARISON WITH THE PREVIOUS BATCH

We kept the previous batch's foundation (MATLAB, MFCC features, energy-based endpointing, a GUI) and replaced the parts that limited its security: the matcher, the login protocol, the noise handling and the evaluation (Tables IX and X).

A. Evaluation Comparison

The previous batch reported closed-set identification: 97.5% of 80 clean test utterances matched the right one of 16 speakers, and 73.81% of 42 real logins succeeded. Closed-set accuracy always names some enrolled speaker, so it cannot show how often an outsider is accepted.

We report verification error rates of both kinds, on recordings held out from the models, with confidence intervals: 0%

TABLE IX
DESIGN COMPARISON WITH THE PREVIOUS BATCH'S SYSTEM

Aspect	Previous batch [1]	This project	What changes
Login protocol	Fixed name + ID; after 3 voice failures, a PIN alone grants access	Roll + name, then a random 3-digit code issued only after Factor 1 passes; no PIN path	A recorded login is no longer enough, and voice cannot be bypassed
What was said	Not checked	DTW on CMVN-normalised MFCC sequences; roll and name must both point to the typed ID; each challenge digit recognised separately	Saying another student's ID is rejected
Who said it	32-codeword VQ per speaker; nearest speaker wins	128-component GMM-UBM with MAP adaptation; claimed owner must rank first and exceed an LLR level	An outsider must sound like the owner, not merely nearest
Speaker check on challenge	No challenge	Separate digit-only GMM-UBM	The person answering the code must also be the owner
Temporal order	Discarded by VQ	Preserved and aligned by DTW	Tolerates speaking-rate changes while checking content
Noise	Pre-emphasis and energy trimming only	3-s live ambient profile $\rightarrow$ power spectral subtraction ($\alpha = 2$, $\beta = 0.02$), per-utterance fallback	Steady room noise is removed before features
Score normalisation	None	Cohort ratio against 5 nearest students; CMVN for content	Common microphone and room shifts cancel
Decision threshold	Minimum distortion; no reject threshold reported	θ_{phrase} learned from 5-fold held-out strangers at a 2% target	Threshold set from data, not by eye
Reported metrics	Closed-set accuracy only	FAR, FRR, HTER, EER bounds, 95% CIs, stage-wise live pass rates	Errors of both kinds are quantified with uncertainty
Enrolled speakers	16 in the reported evaluation	29, each with 17 recordings	Larger, complete cohort
Hardware	Built, with a hardware budget	Proposed fail-closed ESP32 integration (not built)	Barrier opens only on an authenticated, single-use grant
Audio retention	Not stated	Live audio held in RAM only; the log stores scores and decisions	Lower privacy risk

TABLE X
SECURITY SCENARIOS

Situation	Previous batch	This project
Attacker replays a recording of the owner's login	Accepted if the voice matches, since the phrase never changes	Factor 1 may pass, but the 3-digit code is unpredictable (1 in 1000 to guess) and must be spoken in the owner's voice
Owner enters one ID but says another	Not detected	Rejected (observed live: typed 2206154, spoke 2206143's details)
Enrolled classmate claims another student	Depends on which VQ codebook is nearest	Rejected by G3: 0% held-out acceptance; 10 live denials at G3
Unenrolled outsider says their own details	Nearest speaker is always named; no FAR reported	0 of 87 held-out accepted (95% upper bound 4.2%)
Attacker who knows the PIN	Gains access after three deliberate voice failures	No PIN path exists
Spliced or synthetic voice of the owner	No defence	No dedicated defence (Section IX)

FRR (upper bound 1.1%), 0% FAR for unregistered speakers (upper bound 4.2%), 0% classmate impersonation, and a 27.6% worst-case Factor 1 FAR for an outsider who knows whom they resemble. Live sessions are logged with the gate that failed.

The headline numbers of the two projects cannot be subtracted. They measure different tasks, on different speakers and microphones. A fair head-to-head test would run both systems on the same recordings.

B. What Was Not Carried Over

The previous batch also built bicycle check-in/check-out records, slot assignment and an administrator panel. These are application features outside our DSP scope. Its hardware is not reused; instead, Appendix B proposes a barrier interface designed around the security weaknesses identified above.

IX. TECHNOLOGICAL LIMITATIONS

The system stops simple replay and impersonation, but it does not prove a live person is speaking, and its live error rates are not yet measured.

- *No liveness detection:* the challenge uses only ten words. An attacker holding recordings of the owner saying each digit could splice any code, and synthetic speech is not detected.
- *Weak challenge voice model:* each student's digit model is trained on 2–4 s of speech. It caused most denials of sessions that had already passed Factor 1.

- *Same-session enrollment:* all takes of a student were recorded in one sitting. Held-out figures (0% FRR) therefore overstate live performance, where Factor 1 LLRs were 0.39–1.21 against a held-out median of 1.39.
- *Informed outsider:* an unenrolled speaker who knows the details of the student they sound most like passed Factor 1 in 27.6% of held-out trials (CI 18.5–38.2%).
- *Small trial counts:* the zero-error results bound FRR below 1.1% and FAR below 4.2%, not at 0%; live stage rates have CIs up to ± 30 points.
- *Pause-dependent segmentation:* the digits must be separated by roughly 0.25–0.5 s of silence; fluent speech is rejected.
- *Stationary-noise assumption:* one noise profile is used for the whole login. Sudden sounds such as horns or engines starting are not removed, and noise robustness has not been measured systematically.
- *Hand-set gates:* the voice gates (0.30 and 0.20) and the 1.5% digit margin are engineering choices, not calibrated from labelled live trials.
- *Scale and channel:* each login runs DTW against every student's takes, so cost grows with enrollment. Enrollment and use should share one microphone, and the 8-kHz rate discards cues above 4 kHz.
- *Non-cryptographic randomness:* codes come from MATLAB's default pseudo-random generator, adequate for a demonstration but not for deployment; the proposed hardware would move this to the ESP32's hardware RNG.
- *Hardware unbuilt:* Appendix B is a design. Its security and safety properties are untested, and the laptop holding key K remains a trusted component.

A. Departures From the Proposal

Table XI records where the final system differs from the project proposal.

X. Public Health, Safety, Societal, Cultural and Environmental Considerations

The design fails closed and keeps no voice recordings; its main social cost is that a genuine user can be locked out.

Safety: Any failed gate, missing model or segmentation error denies access, and a logging failure cannot change a decision. Because genuine users are sometimes denied, a deployment needs a staffed fallback, such as an attendant who checks photo ID. It must not be a code typed at the same kiosk, which would recreate the previous batch's bypass. A physical barrier adds a second safety duty: it must never close on a vehicle or person, which the proposed design handles with a safety beam and obstruction reversal. All proposed electronics run at 5 V or below.

Health and accessibility: Speaking is contactless, so no shared scanner or keypad is touched. A cold, hoarseness or a speech impairment changes the voice and can cause rejection. These users need the same staffed alternative and must not be disadvantaged by it.

Cultural and linguistic fairness: Each student is matched against their own recorded names and numbers, so any language, accent or name works as long as enrollment and use match. Digits must be spoken in the language used at enrollment.

Privacy: A voice is biometric personal data. Enrollment needs informed consent, and a user must be able to withdraw and have their models deleted. Live audio stays in memory and is discarded; only MFCC templates, model means and a metadata log (IDs, scores, decisions) are stored locally. These files still identify people and must be access-controlled.

Environment: The implemented system is software on an ordinary computer and microphone. The proposed hardware adds only a low-power microcontroller, a servo and a few sensors, all reusable after the project, which limits electronic waste. Toolbox-free MATLAB keeps the software portable to existing machines.

Over-trust: A biometric gate can give a false sense of security. Operators should be told plainly that the system resists replay of a fixed phrase, not spliced or synthetic speech.

XI. Conclusion and Future Work

We turned a fixed-phrase voice identifier into a two-factor voice verifier that checks both the words and the speaker, at both factors. On held-out database trials with 29 students, Factor 1 rejected no genuine logins and accepted no unregistered speakers or impersonating classmates, bounding FRR below 1.1% and FAR below 4.2% at 95% confidence. A random three-digit challenge means a recorded login is no longer enough, and removing the PIN path means voice cannot be skipped.

Live use showed where the system is still weak. Factor 1 and digit recognition worked well (93% of digits correct), but the digit-only voice model denied many sessions that had already passed Factor 1: only 29% of sessions that cleared Factor 1 were granted, and 5 of 29 logged sessions overall. We also proposed, but did not build, a hardware integration in which an ESP32 controller supplies the challenge code from its hardware random-number generator and opens a barrier only on an HMAC-signed, single-use grant from MATLAB, failing closed on any fault. The design and its planned tests are given in Appendix B. A recorded demonstration of the system is available online [17].

The next steps, in order of impact:

1) Record more isolated-digit takes per student, ideally over several days, and retrain the digit voice model.
2) Run a labelled live study across days (genuine users, classmates, outsiders and replayed recordings) to measure true FAR and FRR, and export raw scores for the exact EER and DET curve.
3) Calibrate the Factor 2 gate per student with `CALIBRATE_FACTOR2.m` once enough labelled trials exist.
4) Build the hardware prototype of Appendix B and run the tests in Table XV.

TABLE XI
DEPARTURES FROM THE PROJECT PROPOSAL

Proposal	Final system	Reason
Record 1 s of ambient noise	3 s, validated, reused for all three utterances	A longer, checked reference gives a steadier noise estimate
Spectral floor $\beta \approx 0.02$	$\beta = 0.02$ with over-subtraction $\alpha = 2$	As proposed
50th-order Hamming FIR band-pass, 300–3400 Hz	Not used; band set by the 8-kHz rate and the 100–4000-Hz mel filterbank	Removed during redesign
Replace VQ with DTW	DTW for content; GMM-UBM for speaker identity	In a 27 September live test, VQ ranked the true speaker first in 2 of 12 genuine logins; GMM-UBM did in 11–12 of 12
Name, ID and digits in one utterance	Three separate recordings	Shorter recordings segment more reliably
Threshold from an error-rate crossover	θ_{phrase} from held-out strangers; voice gates hand-set	No labelled live data yet
FAR and FRR	Held-out, same-session only, with 95% CIs and EER bounds	Labelled live trials still needed
Noise robustness with vs. without subtraction	Per-session A/B diagnostic only	Controlled study not yet run
Replay attack tests	Not performed	Planned future work

5) Add a liveness check, for example by challenging with longer or less predictable content, or by detecting playback.
6) Measure accuracy with and without spectral subtraction under recorded garage noise at fixed SNRs.
7) Test the previous batch's system on the same recordings for a fair head-to-head comparison.

APPENDIX A
USER MANUAL

A walkthrough of the login procedure is available as a demonstration video [17].

Requirements: MATLAB with no extra toolboxes, and a microphone. Use the same microphone for enrollment and verification. To choose a specific one, set `cfg.InputDeviceName` in `vsConfig.m` to part of its name.

Folder layout: Keep the three folders in Table XII side by side. A `verification_logs/` folder is created beside them for the session log.

TABLE XII
FOLDER LAYOUT

Folder	Contents
`ParkingSecuritySystem/`	All MATLAB code
`raw_records/`	One folder per student with enrollment WAV files
`database/`	`EnrolledDatabase.mat`, `VerifierModel.mat`, `DigitVoiceModel.mat`

Enrolling students:

1) Create a folder per student in `raw_records/` and add 17 WAV files named `<roll>_<utterance>_<take>.wav`: `roll` and `name` (takes 01–03), `digits` (the sequence 0–9, take 01), and `zero` … `nine` (take 01 each).
2) Run `ENROLL_STUDENTS.m`. It rebuilds the database and both models only if recordings or settings changed; set `FORCE_REBUILD = true` to rebuild anyway.
3) Check the printed held-out error rates and the status summary at the end.

Verifying a user:

1) Run `RUN_ME.m` to open the GUI, or `RUN_ME_COMMAND_WINDOW.m` for a command-window session. Both refuse to start if a model is missing or out of date.
2) Type the claimed 7-digit student ID.
3) Stay silent for 3 s while the room noise is recorded.
4) Say your roll number when prompted, then your full name. Each recording stops by itself after about 0.9 s of silence.
5) If Factor 1 passes, three digits appear. Say them one at a time with a short pause (about 0.25–0.5 s) between digits.
6) The screen shows ACCESS GRANTED or ACCESS DENIED with the reason.

Other tools: See Table XIII.

TABLE XIII
UTILITY COMMANDS

Command	Purpose
`vsIO('miccheck')`	Test the microphone level until it reports GOOD
`ParkingSecuritySystem ('status')`	Show models, students and thresholds
`SHOW_LOG_SUMMARY`	Count granted, denied and per-factor outcomes in the log
`CALIBRATE_FACTOR2`	Supervised per-student calibration of the challenge voice gate (12 genuine + 12 impostor trials)

Troubleshooting: "Found 2 (or 4) speech intervals" means the pauses between digits were too short or a digit was split;

TABLE XIV
PROPOSED COMPONENTS

Function	Example part	Role and interface
Controller	ESP32-WROOM-32 board	RNG, HMAC check, state machine; USB serial to laptop
Actuator	MG996R servo	Lifts a model barrier arm; PWM
Actuator supply	5 V, $\geq$2 A adapter	Powers servo separately; shared ground
Presence sensor	HC-SR04 ultrasonic	Detects a vehicle and wakes the GUI; 5-V echo divided to 3.3 V
Safety sensor	IR break-beam pair	Prevents the arm closing on a vehicle; GPIO
Display	16$\times$2 LCD, I^2C backpack	Shows challenge code and status
Indicators	Red/green LEDs, piezo buzzer	Granted/denied feedback; GPIO
Override	Key switch	Attendant opening, logged; GPIO
Microphone	USB cardioid microphone	Speech input; same unit as enrollment

TABLE XV
PLANNED HARDWARE TESTS (NOT YET PERFORMED)

Test	Procedure	Metric / pass criterion
Actuation	50 scripted GRANT and 50 DENY decisions	Opens on every valid GRANT, never on DENY
Latency	Timestamp grant in MATLAB and servo start	Mean and maximum, ms
Forged grant	GRANT with wrong or missing HMAC	0 openings
Replayed grant	Resend a captured valid GRANT	0 openings
Link loss	Unplug USB or close MATLAB mid-login	Stays closed; status shown
Power cycle	Cut and restore power in each state	Boots to CLOSED
Obstruction	Break safety beam while closing	Arm stops and reopens

repeat with clearer gaps. A "clipped" warning means the input is too loud; lower the Windows input volume and run the mic check.

APPENDIX B
PROPOSED HARDWARE INTEGRATION

This appendix is a design proposal. None of it has been built or tested, and no hardware measurement is reported. Its goal is to connect the MATLAB decision to a physical barrier without creating a new bypass: the barrier must open only on a genuine, fresh grant, and must stay closed on any fault.

A. Architecture

All DSP stays on the laptop, unchanged (Fig. 3). An ESP32 microcontroller acts only on the final decision. Keeping the signal processing in MATLAB avoids porting two 128-component GMMs and DTW against 29 students to a microcontroller, and keeps enrollment and verification on the same USB microphone. MATLAB's built-in `serialport` interface keeps the system toolbox-free.

B. Components

Table XIV lists one possible bill of materials. A cost table with actual local prices should be added once parts are purchased.

C. Authenticated Grant Protocol

A plain "OPEN" serial command would let anyone who plugs a device into the controller's USB port open the barrier without speaking, recreating the PIN bypass in hardware. The proposed exchange prevents this with a secret key K shared by MATLAB and the ESP32 and an HMAC-SHA256 tag [18]:

1) The presence sensor reports a vehicle; MATLAB starts a login.
2) After Factor 1 passes, MATLAB requests a challenge. The ESP32 draws the three digits and a 128-bit nonce N from its hardware random-number generator [19], shows the code on the LCD, and sends both to MATLAB. This also replaces MATLAB's pseudo-random generator for the code.
3) If Factor 2 passes, MATLAB sends $\texttt{GRANT} \,\|\, \text{ID} \,\|\, \text{HMAC}_K(N \,\|\, \text{ID} \,\|\, \texttt{GRANT})$.
4) The ESP32 opens only if the tag is valid, N is the outstanding nonce, and fewer than 30 s have passed since it was issued. N is then discarded, so a captured grant cannot be replayed.
5) A DENY, a timeout or an invalid tag discards N and leaves the barrier closed.

At 115 200 baud (11 520 bytes/s with 10 bits per byte), an 80-byte grant frame takes about 7 ms, negligible beside the speech processing.

D. Fail-Closed Behaviour

The controller boots to CLOSED and returns there after any reset. The hardware watchdog resets a hung controller. If no heartbeat arrives from MATLAB for 5 s, the controller shows "System offline" and accepts no grant until the link recovers. After a valid grant the arm opens, waits until the safety beam has been clear for 3 s, and then closes; it never closes while the beam is broken. On power loss a model arm should be balanced to come to rest closed slowly; a full-size installation would use a commercial barrier unit with its own obstacle detection. The key switch gives the attendant a physical override for genuine users who are denied, as recommended in Section X; it is logged and is not a code typed at the kiosk. Enabling the ESP32's secure boot and flash encryption [19] would stop an attacker with physical access from reading K or reflashing the controller.

E. Planned Hardware Tests

Table XV defines the measurements that would accompany a built prototype.

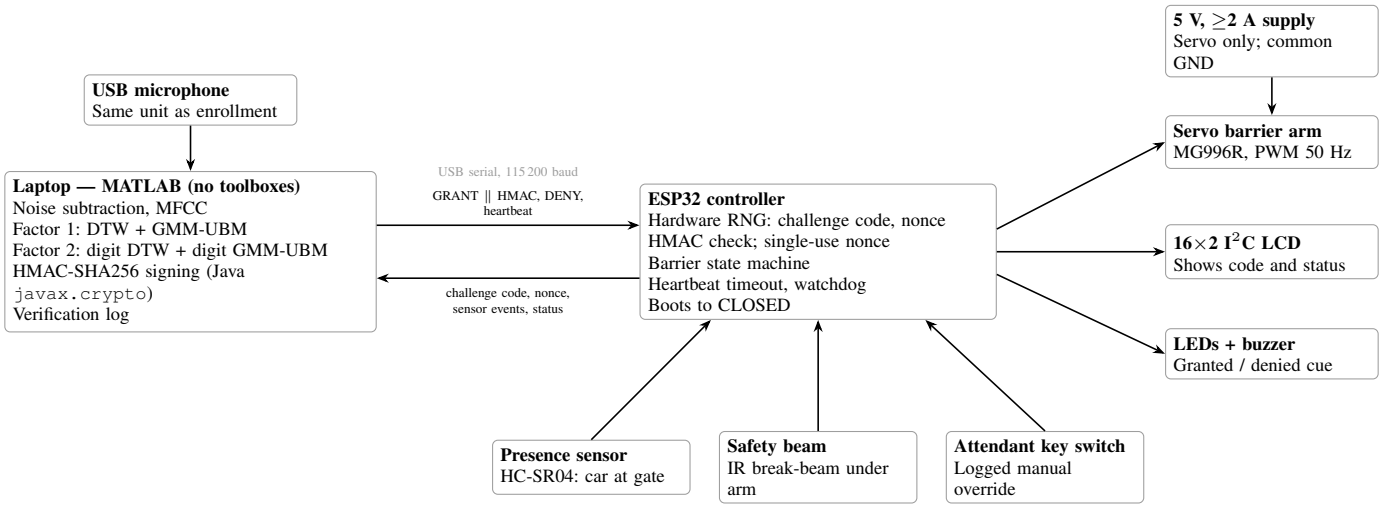


Fig. 3. Proposed hardware integration (design only, not built). MATLAB keeps all signal processing; the ESP32 supplies randomness and moves the barrier only on a valid, single-use signed grant.

Ethics Declaration

We, the members of Group 04, declare that:

1) Every one of the 29 enrolled speakers recorded their samples voluntarily, was told how the recordings would be used, and may ask for their data to be deleted.
2) Recordings and trained models are used only for this course project and are not shared outside the group and the course instructors.
3) All impostor and impersonation tests were performed by consenting participants inside the group's test sessions; no one's voice was used without their knowledge.
4) All algorithms, papers and prior student work we built on are cited in the References, including the previous batch's project [1]. Any code adapted from other sources or groups is identified as such.
5) The results reported here are the system's actual outputs. Limitations and unmeasured quantities are stated rather than estimated, and the hardware design is identified as unbuilt.

Student ID	Name	Signature
2206138	Sabyasachi Dash Nirjhar	
2206143	Mahin Abrar Chowdhury	
2206148	Md. Eftekhar Uddin Khondaker	
2206151	Rumaisa Islam Binte Rubayet	

Teamwork and Individual Contribution

The work was divided into nine packages; each member's role in each is recorded in Table XVI (L = led, C = contributed).

TABLE XVI
Individual Contribution by Work Package

Work package	2206138	2206143	2206148	2206151
Recording protocol and enrollment data				
Pre-processing and noise subtraction				
MFCC extraction and DTW content matching				
GMM-UBM speaker models, Factor 1 gates				
Three-digit challenge				
Threshold training, held-out evaluation				
GUI, live testing, verification log				
Performance analysis and hardware design				
Report, user manual, video presentation				

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